ITAI-1371-Intro to Machine Learning-RT-14274

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Lab 12- Ethics, Fairness, and Bias in ML

Reflective Journal

This lab helped our team understand that evaluating a machine learning model is not only about achieving high accuracy. We learned that a model can perform well overall and still produce very different outcomes for different demographic groups. This made us think more carefully about the ethical responsibility involved in building and evaluating machine learning systems.

In this lab, we used the Adult dataset to train a Logistic Regression model that predicts whether a person's income is greater than $50,000 per year. The model achieved an overall accuracy of 84.61%, which initially appeared to be a good result. However, once we evaluated the model separately for males and females, we discovered an important difference. The model achieved 81.20% accuracy for males and 91.81% accuracy for females. This showed us why overall accuracy can hide important differences between subgroups.

One of the most important lessons from the lab was that higher accuracy does not automatically mean that a model is fair. The female subgroup had higher accuracy, but when we examined the specific types of errors, we found a more complicated situation. The False Positive Rate for males was 10.26%, compared with only 2.81% for females. This means that males were more likely to be incorrectly predicted as having a high income when they actually did not.

At the same time, the False Negative Rate was much higher for females. The model had a False Negative Rate of 37.80% for males and 47.84% for females. This means that women who actually had an income above $50,000 were more likely to be incorrectly classified as having a lower income. This result was especially important to us because it showed that fairness cannot be evaluated by looking at a single metric.

The differences in error rates also helped us understand how machine learning decisions can have real-world consequences. If a similar model were used in lending, a false positive could result in approving a loan for someone whose actual financial situation may not

support it. In a hiring scenario, a false negative could be more concerning because a qualified candidate could be rejected. If one demographic group experiences more false negatives than another, the system could create an unfair disadvantage for that group.

Another important concept we learned was that bias can come from the data itself. The notebook introduced historical bias, measurement bias, and representation bias. Machine learning models learn patterns from past data, and if the data reflects existing inequalities, the model can reproduce those inequalities. The algorithm does not need to intentionally discriminate for unfair outcomes to appear.

We also learned that simply removing a sensitive feature such as sex would not necessarily solve the fairness problem. Other variables, such as occupation, education, marital status, and hours worked, may still be related to gender because of historical or social patterns. These variables can act as proxies, allowing the model to indirectly learn similar information even when the sensitive feature is removed. This helped us understand the concept of proxy bias and why fairness requires deeper analysis than simply deleting one column.

One of the biggest changes in our thinking from this lab was understanding that model evaluation should include questions about who is affected by the model and how. In earlier labs, we mainly focused on metrics such as accuracy, precision, recall, or model performance. In this lab, we learned to ask additional questions: Does the model perform equally well for different groups? Are the errors distributed fairly? Could those errors create disadvantages in a real-world application?

Overall, this lab helped our team see machine learning from a broader perspective. A model can be technically successful while still raising ethical concerns. We now understand that responsible machine learning requires evaluating both performance and fairness before deployment. The most important lesson we gained is that a good model should not only be accurate, but should also be examined carefully to understand how its predictions and errors affect different groups of people.